

PLANWAM: PLANNING-SHAPED FUTURE REPRESENTATIONS FOR END-TO-END AUTONOMOUS DRIVING

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ABSTRACT

World models in end-to-end autonomous driving predict future scene evolution to provide foresight for trajectory planning. Existing methods mainly study how to predict the future and how to use it, but less often ask which future representation is actually most useful for planning. To this end, we propose PlanWAM, a Planning-Shaped World-Action Model. The key idea is to let the planning task shape the future-state representation, so that it retains the information most useful for planning. A latent world model then predicts this planning-oriented future latent representation from historical observations and uses it for planning, enabling foresighted planning. Specifically, we first use a Temporal Register Pyramid to compress multi-frame historical information in a recency-aware manner, learning a compact historical representation oriented toward future reasoning and planning. We then introduce a privileged future posterior that observes ground-truth future frames, and shape its future latent representation with trajectory-planning objectives to obtain a planning-oriented future latent representation. Hindsight-to-Foresight Distillation trains a causal prior that depends only on history to predict this future latent representation. The predicted future latent representation serves as planning context and guides trajectory generation and selection. PlanWAM achieves 93.8 PDMS / 90.9 EPDMS on NAVSIM-v1/v2 navtest and reaches 38.7 HD-Score on closed-loop HUGSIM in a zero-shot setting, demonstrating leading planning performance across both open-loop and closed-loop evaluations. Extensive experiments further demonstrate that planning-shaped future representations provide an effective and deployable form of foresight for world-action models.

1 INTRODUCTION

End-to-end autonomous driving uses a unified model to plan future trajectories directly from raw sensor observations, so that scene representations can be learned for the final planning objective Codevilla et al. (2018); Hu et al. (2023); Jiang et al. (2023); Liao et al. (2025). Reliable driving, however, depends on more than the current observation. A planner must also use history to identify motion trends and interactions among traffic participants, and to infer possible future scene evolution. World models have therefore become an important research direction for end-to-end autonomous driving Wang et al. (2024a); Zheng et al. (2024); Zhou et al. (2025a). By modeling how the scene evolves over time and with vehicle behavior, they supply future information to the planner, shifting trajectory planning from reacting to the current scene toward foresighted planning.

Existing driving world models have gradually changed how they represent and use future information, moving from explicit future reconstruction to latent prediction and then to future-conditioned planning (Fig. 1). Early methods, such as Epona Zhang et al. (2025) and PWM Zhao et al. (2025), learn environment dynamics by generating future images, videos, or dense scene states, but they often have to model substantial visual detail that is only weakly related to planning, at a high generation cost. Later methods, such as LAW Li et al. (2025a) and Latent-WAM Wang et al. (2026b), turn to latent world modeling and predict future states in a compact feature space. Their future representations mainly serve as prediction targets or auxiliary supervision, so the effect on planning remains indirect. Recent methods, such as DriveFuture Hong et al. (2026) and WorldDrive Gui et al. (2026), further introduce the predicted future latent explicitly into the trajectory planner, so that future information can participate in planning directly. Even so, most existing methods focus on how to predict the future and how to use it, and less often study a more basic question: which future representation

is most useful for planning? Existing future representations are usually defined by generic feature prediction, visual reconstruction, or a preset state objective, and may therefore be mismatched with the downstream planning objective: a future representation that can reconstruct the scene does not necessarily retain the key information that affects trajectory decisions.

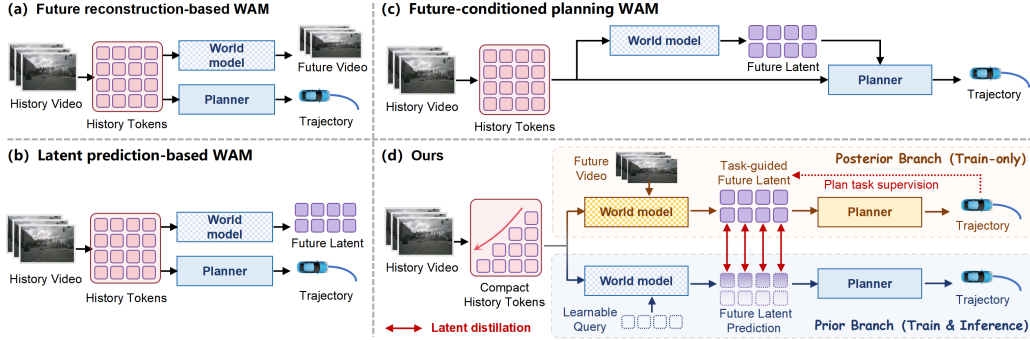


Figure 1: **Four future-modeling paradigms in driving world-action models (WAM).** (a) Reconstruction. (b) Latent prediction. (c) Future-conditioned planning. (d) PlanWAM (ours).

We reformulate future modeling in end-to-end driving as a planning-oriented predictive representation learning problem: a world model must not only predict the future, but also learn which future information is worth predicting. For autonomous driving, an effective representation need not fully retain historical observations or reconstruct the future scene, but should focus on information relevant to trajectory planning. The historical representation should distill planning-relevant dynamic cues from redundant multi-frame, multi-view observations to support future reasoning and planning, while the future representation should be directly shaped by trajectory planning objectives to preferentially encode decision-relevant future information. Furthermore, this planning-oriented future representation should be predictable from historical observations and serve as foresight for trajectory planning. From this perspective, we unify historical modeling, future prediction, and trajectory planning into a planning-oriented predictive representation learning framework, where planning defines what future information matters, the future shapes what historical information should be preserved, and the predicted future representation in turn improves planning.

Building on this perspective, we propose PlanWAM, a Planning-Shaped World-Action Model, as illustrated in Fig. 1(d). First, we compress multi-frame, multi-view historical observations in a recency-aware manner and learn a compact historical memory under joint future-prediction and planning supervision, preserving key dynamic cues for future reasoning. We then introduce a training-only privileged future posterior that leverages ground-truth future observations to construct a future latent representation, whose representation space is shaped by trajectory planning objectives to obtain a planning-oriented future latent representation. Since future observations are unavailable at deployment, Hindsight-to-Foresight Distillation trains a causal prior conditioned only on history to predict this representation, thereby converting privileged hindsight during training into deployable foresight at inference time. The predicted future representation is fused with the historical memory as a foresight planning context to guide trajectory generation and selection, enabling Foresight-Conditioned Planning. Our main contributions are as follows:

- We reformulate future modeling in end-to-end driving as a planning-oriented predictive representation learning problem, where the future representation is shaped directly by trajectory planning objectives rather than predefined reconstruction or generic prediction objectives, so that it preferentially encodes decision-relevant future information.
- We propose PlanWAM, which extracts a compact history representation for future reasoning, learns planning-shaped future latents from ground-truth future observations and planning objectives, and uses Hindsight-to-Foresight Distillation to predict this representation from history as foresight for trajectory generation and selection.
- PlanWAM achieves 93.8 PDMS / 90.9 EPDMS on NAVSIM-v1/v2 navtest and 38.7 HD-Score in zero-shot closed-loop evaluation on HUGSIM. Further analyses show that planning-shaped future representations improve trajectory generation and selection while reducing safety-critical failures, including collisions and low-TTC events.

2 RELATED WORK

2.1 END-TO-END AUTONOMOUS DRIVING

End-to-end autonomous driving uses a unified model to plan future trajectories directly from raw sensor observations, thereby reducing information loss and cascading errors in traditional modular systems. Early methods, such as UniAD Hu et al. (2023), VAD Jiang et al. (2023), SparseDrive Sun et al. (2025), and TransFuser Chitta et al. (2022), used structured scene representations and multi-task learning to jointly model perception, prediction, and planning. Subsequent methods, such as iPad Guo et al. (2026), DrivoR Kirby et al. (2026), DiffusionDrive Liao et al. (2025), and GoalFlow Xing et al. (2025), adopt multimodal and generative trajectory planning to capture diverse and uncertain driving behaviors. More recent methods, such as ReCogDrive Li et al. (2026c), AutoVLA Zhou et al. (2025b), Orion Fu et al. (2025), and Qwen-Drive-1.0 Zhou et al. (2026b), introduce vision-language and vision-language-action models to strengthen semantic understanding and reasoning. Although these methods keep improving scene representation and trajectory planning, they still decide mainly from current or historical observations and make limited use of future scene evolution. In contrast, PlanWAM uses a world model to predict a future representation useful for planning, and guides trajectory generation and evaluation with that foresight.

2.2 WORLD MODELS FOR AUTONOMOUS DRIVING

World models provide foresight for end-to-end driving by modeling how the scene evolves over time and under vehicle behavior. Existing driving world models fall into two lines. One line, such as DriveDreamer Wang et al. (2024a), Drive-WM Wang et al. (2024b), Vista Gao et al. (2024), Epona Zhang et al. (2025), and PWM Zhao et al. (2025), focuses on generating and simulating the future real world. These methods learn traffic evolution by predicting images, videos, or structured scene states such as BEV, and mainly study generation quality, temporal consistency, action controllability, and long-horizon prediction. The other line, such as LAW Li et al. (2025a), LatentWAM Wang et al. (2026b), World4Drive Zheng et al. (2025), DriveFuture Hong et al. (2026), and WorldDrive Gui et al. (2026), focuses on the role of world modeling in the driving task itself, learning dynamic scene representations by predicting future features or latent states and further assisting or directly participating in trajectory planning. These studies show that future modeling can provide a useful dynamic prior for end-to-end driving. Existing work, however, mainly studies how to predict future features more effectively and how future information is used, and less often asks, from the planning objective itself, what the future representation should encode. In contrast, PlanWAM studies which future representation is most useful for planning and shapes the future latent representation directly with planning objectives, so that the latent preferentially extracts the key future information relevant to planning decisions.

3 METHOD

3.1 PROBLEM FORMULATION AND FRAMEWORK OVERVIEW

Let $v_{t-n:t}$ denote the historical multi-view camera observations, $c_{t-n:t}$ the corresponding ego states and navigation information, and $\tau_{1:T}$ the future ego trajectory to be planned. A conventional end-to-end planner directly models:

$$p(\tau_{1:T} \mid v_{t-n:t}, c_{t-n:t}).$$

To use future scene evolution explicitly, future-conditioned world-action models introduce a future-state representation z_t^+ between historical observations and trajectory planning, and factorize the process as:

$$p(\tau_{1:T}, z_t^+ \mid v_{t-n:t}, c_{t-n:t}) = p(z_t^+ \mid v_{t-n:t}, c_{t-n:t}) p(\tau_{1:T} \mid v_{t-n:t}, c_{t-n:t}, z_t^+). \quad (1)$$

This factorization lets future information participate in planning, but it also raises a key question: *how should z_t^+ be defined and constructed so that it is more beneficial for trajectory planning?* Existing methods typically define the future representation by generic feature prediction, visual reconstruction, or a predefined state target, but such a representation is not necessarily the most useful for planning. To this end, we propose PlanWAM, as illustrated in Fig. 2.

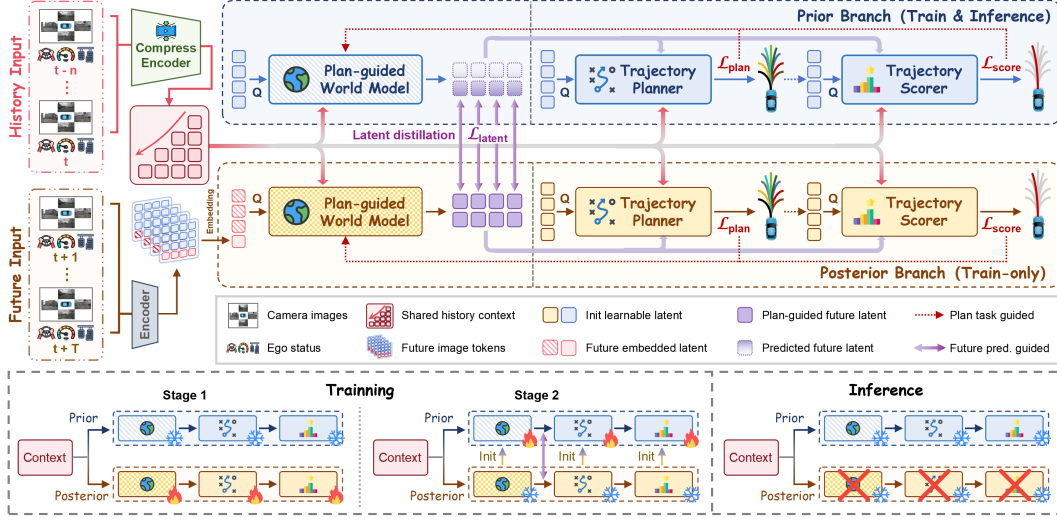


Figure 2: **Overview of PlanWAM.** Historical multi-view observations are compressed into a compact representation for future reasoning. The privileged posterior branch uses ground-truth future observations and planning objectives to learn planning-shaped future latents, which are distilled into a causal prior conditioned only on history. The predicted future latents then serve as foresight for trajectory generation and selection. The posterior is training-only; inference uses only the prior.

We first use a history compression module E_η to compress multi-frame, multi-view historical observations in a recency-aware manner, yielding a compact historical representation:

$$h_t = E_\eta(v_{t-n:t}, c_{t-n:t}). \quad (2)$$

During training, we further introduce a privileged future posterior P_ϕ that accesses ground-truth future observations and constructs a future latent representation:

$$z_t^+ = P_\phi(h_t, v_{t+1:t+T}, c_{t+1:t+T}). \quad (3)$$

Rather than prescribing the semantics of z_t^+ with an auxiliary reconstruction or generic prediction objective, we condition trajectory generation and selection directly on z_t^+ :

$$\tau_{1:T} \sim \pi_\phi(h_t, z_t^+). \quad (4)$$

The future latent space is therefore shaped jointly by the trajectory generation and trajectory selection objectives, so that z_t^+ preferentially encodes future information that affects planning. We refer to this representation as a *planning-shaped future representation*.

However, the future observations $v_{t+1:t+T}$ and $c_{t+1:t+T}$ are unavailable at deployment. PlanWAM therefore learns a causal prior P_θ that depends only on the historical memory and is trained by Hindsight-to-Foresight Distillation to predict z_t^+ :

$$\hat{z}_t^+ = P_\theta(h_t), \quad \hat{z}_t^+ \approx z_t^+. \quad (5)$$

The predicted planning-shaped future representation $\hat{z}_t^+$ is fused with the compact historical representation h_t as a foresight planning context to guide trajectory generation and selection:

$$\tau_{1:T} \sim \pi_\theta(h_t, \hat{z}_t^+). \quad (6)$$

At inference, the planning process of PlanWAM can be expressed as:

$$p(\tau_{1:T}, \hat{z}_t^+ | v_{t-n:t}, c_{t-n:t}) = p_\theta(\hat{z}_t^+ | h_t) p_\theta(\tau_{1:T} | h_t, \hat{z}_t^+). \quad (7)$$

3.2 RECENCY-AWARE HISTORICAL REPRESENTATION

Multi-frame, multi-view historical observations contain substantial temporal redundancy, and information from different temporal ranges has different value for planning. To this end, we propose a

Temporal Register Pyramid that compresses historical scene registers level by level in a recency-aware manner, retaining, under a limited token budget, the dynamic information relevant to future reasoning and trajectory planning, as illustrated in Fig. 3.

Given the historical observations $v_{t-n:t}$ and ego context $c_{t-n:t}$, a shared visual encoder maps the k -th camera observation v_i^k at time i to visual tokens and further extracts scene registers r_i^k ; the ego state and navigation information c_i are encoded by an MLP into an ego feature e_i :

$$x_i^k = F_{\text{vis}}(v_i^k), \quad r_i^k = F_{\text{reg}}(x_i^k), \quad e_i = F_{\text{ego}}(c_i). \quad (8)$$

We partition the history into L temporal levels $\{\mathcal{G}_l\}_{l=1}^L$ according to temporal distance, and assign different query budgets: recent levels use more learnable queries, whereas more distant history is compressed more strongly. For the l -th level, the queries $Q^{(l)}$ are fused with the ego feature as well as time, camera, and compression-level embeddings; the scene registers at the current time are aggregated into a shared current-scene context $\bar{r}_t$:

$$\tilde{Q}^{(l)} = Q^{(l)} + E_{\text{time}}^{(l)} + E_{\text{cam}}^{(l)} + E_{\text{level}}^{(l)} + E_{\text{ego}}^{(l)} + \bar{r}_t. \quad (9)$$

We then apply a Q-Former to perform query-based compression of the historical information at each level, using $\tilde{Q}^{(l)}$ as Q and the scene registers $R^{(l)}$ of the corresponding level as K and V :

$$h_t^{(l)} = \text{QFormer}(Q = \tilde{Q}^{(l)}, K = R^{(l)}, V = R^{(l)}), \quad R^{(l)} = \text{Concat}_{i \in \mathcal{G}_l, k} r_i^k. \quad (10)$$

The compressed tokens from all levels constitute the compact historical representation h_t . We do not use an additional historical reconstruction objective; instead, h_t is jointly optimized by the subsequent future-latent distillation objective $\mathcal{L}_{\text{latent}}$ and the trajectory-planning objective $\mathcal{L}_{\text{task}}$, so that it focuses on the information required for future prediction and trajectory planning.

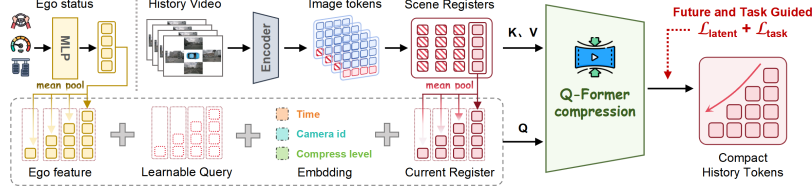


Figure 3: Temporal Register Pyramid for Recency-Aware Historical Representation.

3.3 PLANNING-SHAPED FUTURE REPRESENTATION

Given the compact historical representation h_t from Sec. 3.2, we construct a privileged future posterior during training using future observations. For the future observations $v_{t+1:t+T}$, a shared visual encoder extracts scene registers r_i^k for each time and camera in the same way as for historical observations, and adds the corresponding future-time embedding $E_{\text{time}}^{\text{fut}}$ to form future queries:

$$f_t^+ = \text{Concat}_{i,k} [r_i^k + E_{\text{time}}^{\text{fut}}(i - t)], \quad i \in \{t + 1, \dots, t + T\}. \quad (11)$$

Rather than using randomly initialized learnable queries, the privileged posterior P_ϕ takes f_t^+ formed from ground-truth future scene registers directly as queries, and uses the historical representation h_t as context. The posterior world model $\text{WM}_\phi^{\text{post}}$ aggregates historical evidence and ground-truth future information to obtain the posterior future latent z_t^+ :

$$z_t^+ = P_\phi(h_t, f_t^+) = \text{WM}_\phi^{\text{post}}(Q = f_t^+, K = h_t, V = h_t). \quad (12)$$

We do not apply an additional generic reconstruction objective to the future latent z_t^+ . Instead, we concatenate z_t^+ with the historical representation h_t to form a posterior planning context m_t^+ , adding type embeddings before concatenation to distinguish historical evidence from future context. The resulting m_t^+ is used for both trajectory planner and trajectory scorer:

$$m_t^+ = \text{Concat}(h_t + E_{\text{type}}^{\text{hist}}, z_t^+ + E_{\text{type}}^{\text{fut}}), \quad \tau_{1:T}^+ = G_{\text{plan}}(m_t^+), \quad s^+ = G_{\text{score}}(\tau_{1:T}^+, m_t^+). \quad (13)$$

The posterior branch is supervised by the trajectory generation and trajectory selection objectives:

$$\mathcal{L}_{\text{post}} = \mathcal{L}_{\text{plan}} + \mathcal{L}_{\text{score}}. \quad (14)$$

The planning losses can therefore back-propagate through the planner and the scorer to shape z_t^+ directly. The limited future latent capacity thus preferentially retains future scene information that affects trajectory generation and selection. We refer to this z_t^+ as a *planning-shaped future representation*. The next section uses it as a privileged target to train a history-only causal prior.

3.4 HINDSIGHT-TO-FORESIGHT PREDICTION AND PLANNING

The planning-shaped future representation z_t^+ in Sec. 3.3 relies on future observations and is therefore unavailable at deployment. We thus construct a causal prior P_θ conditioned solely on the historical representation h_t to predict the corresponding future latent. Unlike the posterior, which directly uses ground-truth future scene registers as queries, the prior uses learnable future queries Q^{prior} . To establish a consistent token-wise correspondence between the prior and posterior latents, we arrange the prior queries in the same order over future times, camera views, and register slots, and augment them with the corresponding structured embeddings:

$$Q_{i,k,j}^{\text{prior}} = Q_{0,i,k,j}^{\text{prior}} + E_{\text{time}}^{\text{fut}}(i-t) + E_{\text{cam}}(k) + E_{\text{slot}}(j), \quad i \in \{t+1, \dots, t+T\}. \quad (15)$$

where i , k , and j index the future time, camera view, and scene-register slot, respectively. The prior world model $\text{WM}_\theta^{\text{prior}}$ then predicts the planning-shaped future representation $\hat{z}_t^+$ from h_t alone:

$$\hat{z}_t^+ = P_\theta(h_t) = \text{WM}_\theta^{\text{prior}}(Q = Q^{\text{prior}}, K = h_t, V = h_t). \quad (16)$$

To transfer the future information captured by the privileged posterior to the causal prior, we propose Hindsight-to-Foresight Distillation, which directly aligns the prior prediction $\hat{z}_t^+$ with the posterior target z_t^+ . Through this alignment, the causal prior learns to infer from historical dynamics the planning-shaped representation that the posterior derives from ground-truth future observations, thereby distilling privileged hindsight into causal foresight:

$$\mathcal{L}_{\text{latent}} = \frac{1}{N_z d} \|\hat{z}_t^+ - \text{sg}(z_t^+)\|_F^2. \quad (17)$$

where $\text{sg}(\cdot)$ denotes stop-gradient, $\|\cdot\|_F$ denotes the Frobenius norm, and N_z and d denote the number of tokens in the future latent and the feature dimension of each token, respectively. After adding the corresponding type embeddings, we concatenate the predicted $\hat{z}_t^+$ with the historical representation h_t to form a prior planning context. This context is provided to both the trajectory planner and scorer, yielding the following Foresight-Conditioned Planning formulation:

$$m_t^- = \text{Concat}(h_t + E_{\text{type}}^{\text{hist}}, \hat{z}_t^+ + E_{\text{type}}^{\text{fut}}), \quad \tau_{1:T} = G_{\text{plan}}(m_t^-), \quad s = G_{\text{score}}(\tau_{1:T}, m_t^-). \quad (18)$$

3.5 TRAINING AND INFERENCE

Training. PlanWAM is trained in two stages. The first stage optimizes only the privileged posterior branch, which learns the planning-shaped future representation from ground-truth future observations under trajectory generation, trajectory selection, and lightweight regularization objectives:

$$\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{plan}} + \mathcal{L}_{\text{score}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}. \quad (19)$$

where $\mathcal{L}_{\text{reg}}$ includes diversity regularization to mitigate trajectory mode collapse. In the second stage, we freeze the posterior and use its checkpoint to initialize structurally compatible parameters in the prior. We then fine-tune the causal prior with a lower learning rate using the objective:

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{task}} + \lambda_{\text{latent}} \mathcal{L}_{\text{latent}}. \quad (20)$$

Inference. At inference, we remove the privileged posterior and retain only the causal prior. Historical observations are first encoded into h_t , after which the prior predicts the planning-shaped future representation $\hat{z}_t^+$. This representation serves as foresight and is combined with h_t for trajectory planner and scorer. Consequently, the entire inference process relies solely on observable history.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets and benchmarks. We evaluate PlanWAM on NAVSIM-v1 Dauner et al. (2024), NAVSIM-v2 Cao et al. (2025), and HUGSIM Zhou et al. (2026a) to assess both planning performance and closed-loop generalization. Models are trained on NAVSIM *navtrain* (103,288 scenes) and evaluated on *navtest* (12,146 scenes). NAVSIM-v1 follows the official evaluation protocol and uses PDM Score (PDMS) as the primary metric; NAVSIM-v2 uses the Extended PDM Score (EPDMS). We further perform zero-shot closed-loop evaluation on the high-fidelity closed-loop simulator HUGSIM, deploying the trained model directly without HUGSIM-specific fine-tuning, and report HUGSIM Driving Score (HD-Score) and Route Completion (RC) to assess closed-loop driving performance and cross-environment generalization.

Implementation details. PlanWAM takes $H=4$ historical frames from four cameras (front, left, right, and rear), while the training-only posterior observes future multi-view frames at 1-s intervals over a 4-s horizon. We use a DINOv2 ViT-S backbone with frozen base weights and rank-32 LoRA Hu et al. (2022), extracting 16 scene registers per camera-frame. The Temporal Register Pyramid assigns $[4, 4, 8, 16]$ queries per camera across four temporal levels, producing a 128-token historical representation h_t . Four future times, four cameras, and 16 registers per camera yield $N_z=256$ posterior future-latent tokens. Over the same horizon, the trajectory planner generates $K=64$ candidate trajectories of 8 waypoints, and the trajectory scorer ranks them for final selection. Both stages use AdamW Loshchilov & Hutter (2019) and are trained on 16 NVIDIA A800 GPUs with a per-GPU batch size of 8. The learning rates for Stages 1 and 2 are 1×10^{-4} and 1×10^{-5} , respectively, and the latent-alignment weight is set to $\lambda_{\text{latent}}=0.05$. Further implementation and evaluation details are given in Appendix A.

4.2 MAIN RESULTS

Open-loop Planning on NAVSIM. Tables 2 and 1 report the planning performance on the NAVSIM-v1/v2 `navtest` benchmarks. PlanWAM achieves 93.8 PDMS on NAVSIM-v1, outperforming all learned methods compared and improving the DrivoR baseline by 0.7 points; compared with the strongest prior world-model-based planner, DriveFuture, the gain reaches 3.1 points. On the more challenging NAVSIM-v2 benchmark, PlanWAM further achieves 90.9 EPDMS, exceeding the previous strongest learned result by 1.0 point and the reported Human Agent baseline by 0.6 points. PlanWAM also obtains the best learned NC of 99.0 and TTC of 98.5, with stable gains on safety-related planning metrics. These results show that planning-shaped future representations provide effective foresight for trajectory planning on both NAVSIM benchmarks.

Table 1: **Performance on the NAVSIM-v2 `navtest` benchmark.** The best learned result in each column is shown in **bold**.

Method	Reference	NC $\uparrow$	DAC $\uparrow$	DDC $\uparrow$	TLC $\uparrow$	EP $\uparrow$	TTC $\uparrow$	LK $\uparrow$	HC $\uparrow$	EC $\uparrow$	EPDMS $\uparrow$
Human Agent	-	100.0	100.0	99.8	100.0	87.4	100.0	100.0	98.1	90.1	90.3
<i>E2E-based planners</i>											
TransFuser Chitta et al. (2022)	TPAMI'22	96.9	89.9	97.8	99.7	87.1	95.4	92.7	98.3	87.2	76.7
DiffusionDrive Liao et al. (2025)	CVPR'25	98.2	95.9	99.4	99.8	87.5	97.3	96.8	98.3	87.7	84.5
MeanFuser Wang et al. (2026a)	CVPR'26	98.3	97.2	99.6	99.8	87.6	97.4	97.3	98.3	88.2	89.5
<i>VLM-based planners</i>											
ReCogDrive* Li et al. (2026c)	ICLR'26	98.3	95.2	99.5	99.8	87.1	97.5	96.6	98.3	86.5	83.6
SGDrive Li et al. (2026a)	CVPR'26	98.6	94.3	99.5	99.9	86.0	97.9	96.1	98.3	85.9	86.2
ExploreVLA Sheng et al. (2026)	ECCV'26	98.8	96.2	99.6	99.8	87.1	98.2	97.8	98.3	86.8	88.8
<i>World-model-based planners</i>											
DriveVLA-W0 Li et al. (2026b)	ICLR'26	98.5	99.1	98.0	99.7	86.4	98.1	93.2	97.9	58.9	86.1
DriveLaW Xia et al. (2026)	CVPR'26	98.7	96.9	99.6	99.8	87.5	98.3	97.6	98.4	77.4	88.6
DreamerAD Yang et al. (2026)	ECCV'26	98.0	97.2	99.5	99.8	87.8	97.4	97.5	98.3	72.4	87.7
Latent-WAM Wang et al. (2026b)	arXiv'26	98.1	97.3	99.6	99.8	87.7	97.3	97.6	98.1	87.3	89.3
DriveFuture Hong et al. (2026)	arXiv'26	98.8	99.1	99.6	99.9	86.6	98.4	96.4	98.3	74.8	89.9
PlanWAM (ours)	-	99.0	99.2	99.8	99.6	90.8	98.5	96.5	98.1	70.5	90.9

Table 2: **Performance on the NAVSIM-v1 `navtest` benchmark.**

Method	NC $\uparrow$	DAC $\uparrow$	TTC $\uparrow$	C $\uparrow$	EP $\uparrow$	PDMS $\uparrow$
Human Agent	100	100	100	99.9	87.5	94.8
<i>E2E-based planners</i>						
DiffusionDrive Liao et al. (2025)	98.2	96.2	94.7	100	82.2	88.1
MeanFuser Wang et al. (2026a)	98.6	97.0	95.0	100	82.8	89.0
iPad Guo et al. (2026)	98.6	98.3	94.9	100	88.0	91.7
DrivoR Kirby et al. (2026)	98.9	98.3	96.2	100	89.1	93.1
<i>VLM-based planners</i>						
AutoVLA Zhou et al. (2025b)	98.4	95.6	98.0	99.9	81.9	89.1
ReCogDrive Li et al. (2026c)	97.9	97.3	94.9	100	87.3	90.8
ExploreVLA Sheng et al. (2026)	98.8	98.4	96.5	99.9	83.5	90.4
SGDrive Li et al. (2026a)	98.6	97.8	96.2	100	85.8	91.1
<i>World-model-based planners</i>						
WoTE Li et al. (2025b)	98.5	96.8	94.9	99.9	81.9	88.3
Epona Zhang et al. (2025)	97.9	95.1	93.8	99.9	80.4	86.2
DreamerAD Yang et al. (2026)	98.0	97.2	94.3	100	83.1	88.7
DriveLaW Xia et al. (2026)	99.0	97.1	96.7	100	81.3	89.1
DriveVLA-W0 Li et al. (2026b)	98.7	99.1	95.3	99.3	83.3	90.2
DriveFuture Hong et al. (2026)	98.8	99.1	95.4	100	84.2	90.7
PlanWAM (ours)	98.9	98.8	96.3	100	91.0	93.8

Table 3: **Closed-loop comparison on HUGSIM.** E, M, H, and X denote Easy, Medium, Hard, and Extreme.

Method	E	M	H	X	Avg.
UniAD Hu et al. (2023)	58.6	41.2	40.4	26.0	40.6
VAD Jiang et al. (2023)	38.7	27.0	25.5	23.0	27.9
LTF Chitta et al. (2022)	68.4	40.7	36.9	25.5	41.4
GTRS-Dense Li et al. (2025c)	64.2	50.0	20.7	22.3	38.0
Latent-WAM Wang et al. (2026b)	84.2	42.5	30.6	35.5	45.9
PlanWAM (ours)	87.6	48.3	51.9	37.2	53.7
(a) Route Completion					
Method	E	M	H	X	Avg.
UniAD Hu et al. (2023)	48.7	29.5	27.3	14.3	28.9
VAD Jiang et al. (2023)	24.3	9.9	10.4	8.2	12.3
LTF Chitta et al. (2022)	52.8	24.6	19.8	8.1	24.8
GTRS-Dense Li et al. (2025c)	55.5	39.0	11.7	14.3	28.6
Latent-WAM Wang et al. (2026b)	72.5	24.0	12.2	18.1	28.9
PlanWAM (ours)	75.3	32.9	35.6	22.0	38.7
(b) HD-Score					

Zero-shot Closed-loop Generalization on HUGSIM. Table 3 reports zero-shot closed-loop results on HUGSIM without simulator-specific fine-tuning. PlanWAM achieves 53.7 average RC and 38.7 HD-Score, outperforming the strongest baselines by 7.8 and 9.8 points, respectively. The gains remain pronounced on the Hard / Extreme subsets, where PlanWAM reaches 51.9 / 37.2 RC and 35.6 / 22.0 HD-Score, respectively. These results demonstrate strong zero-shot closed-loop generalization across varying scenario difficulty.

Qualitative Planning Results. Fig. 4 presents representative qualitative results on NAVSIM and HUGSIM. On NAVSIM, compared with DrivoR, PlanWAM produces trajectories that better account for surrounding vehicles and avoid unsafe interactions in challenging traffic scenarios. On HUGSIM, the model transfers zero-shot across KITTI-360, PandaSet, nuScenes, and Waymo, while maintaining stable closed-loop planning in diverse traffic conditions. These examples qualitatively support the consistent gains observed in the quantitative evaluation.

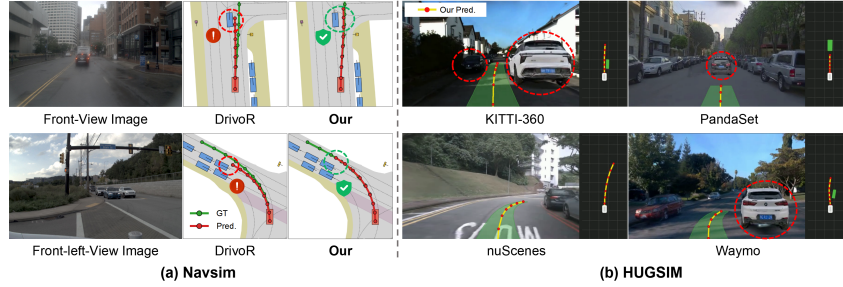


Figure 4: **Qualitative planning results.** (a) Comparison with DrivoR on representative NAVSIM scenarios. (b) Zero-shot closed-loop rollouts on HUGSIM across different driving domains.

4.3 ABLATION STUDIES

We conduct systematic ablations on NAVSIM-v1 `navtest` to analyze the role of each PlanWAM component. Every run follows the full training procedure and changes only the module under study.

Overall Component Evaluation. We add the three core components in sequence, the Temporal Register Pyramid (TRP), Hindsight-to-Foresight Distillation (H2F), and Foresight-Conditioned Planning (FCP), as reported in Table 4. From the base model, TRP raises PDMS from 93.1 to 93.4, showing that a recency-aware historical representation extracts more useful dynamics from multi-frame observations. Adding H2F further reaches 93.5, so learning a future latent from privileged future information supplies additional foresight. FCP then uses the predicted future latent as planning context for trajectory generation and selection, and the model reaches 93.8 PDMS.

Table 4: **Overall component ablation of PlanWAM.**

ID	TRP	H2F	FCP	NC↑	DAC↑	TTC↑	C↑	EP↑	PDMS↑
0	×	×	×	98.3	98.9	94.6	100	90.5	93.1
1	✓	×	×	98.8	98.8	95.3	100	90.9	93.4
2	✓	✓	×	98.6	98.6	95.5	100	91.5	93.5
3	✓	✓	✓	98.9	98.8	96.3	100	91.0	93.8

Future Representation Learning and Utilization. Table 5 shows that generic future-feature supervision reaches only 93.5 PDMS, so a generic prediction target does not by itself yield planning-relevant information. Shaping the future latent directly with the trajectory planning objectives raises performance to 93.8 PDMS. Table 6 then varies where the predicted latent is used. Auxiliary supervision or conditioning only trajectory generation brings little gain, whereas using it for trajectory selection reaches 93.7 PDMS. Conditioning both generation and selection attains 93.8 PDMS.

Historical Representation Compression and Future Distillation. Table 7 shows that, under the same 128-token budget, the recency-aware allocation [4, 4, 8, 16] reaches 93.8 PDMS, above both the uniform allocation and the reversed allocation. Raising the token count to 256 yields only 93.6 PDMS, so how the queries are allocated matters more than enlarging the representation. Table 8

shows that a prior alone has a latent similarity of 0.22. Adding latent distillation raises it to 0.92 and PDMS to 93.7. With posterior initialization, similarity reaches 0.93 and PDMS 93.8.

Table 5: Ablation of Future Representation.

Future representation	Supervision	PDMS $\uparrow$
Null future	—	93.4
Generic future latent	$\mathcal{L}_{\text{feature}}$	93.5
Planning-shaped latent (ours)	$\mathcal{L}_{\text{plan}} + \mathcal{L}_{\text{score}}$	93.8

$\mathcal{L}_{\text{feature}}$: reconstruction of future scene registers.

Table 6: Ablation of Future Latent Usage.

Future latent usage	Generation	Selection	PDMS $\uparrow$
Auxiliary supervision only	×	×	93.5
Generation only	✓	×	93.5
Selection only	×	✓	93.7
Generation + Selection	✓	✓	93.8

Table 7: Ablation of Temporal Allocation.

Query allocation	Tokens	Latency (ms) $\downarrow$	PDMS $\uparrow$
[16, 16, 16, 16]	256	380	93.6
[8, 8, 8, 8]	128	343	93.4
[16, 8, 4, 4]	128	346	92.9
[4, 4, 8, 16]	128	342	93.8
[2, 4, 8, 16]	120	335	93.6

Inference latency on one NVIDIA A800, batch size 1, fp32.

Table 8: Ablation of future latent distillation.

Setting	Posterior init.	$\mathcal{L}_{\text{latent}}$	Latent sim. $\uparrow$	PDMS $\uparrow$
Prior-only	×	×	0.22	93.4
+ Posterior init.	✓	×	0.46	93.5
+ H2F distillation	×	✓	0.92	93.7
Init. + H2F (ours)	✓	✓	0.93	93.8
Privileged posterior (oracle)	—	—	—	95.0

Latent sim.: mean cosine similarity between $\hat{z}_t^+$ and z_t^+ .

4.4 ANALYSIS OF FAILURE REDUCTION

To further understand how planning-oriented foresight improves planning performance, we analyze the PDMS= 0 failure cases on NAVSIM-v1 navtest. As shown in Fig. 5(a), safety-related failures decrease markedly after introducing PlanWAM. In particular, No at-fault Collision (NC) and Time-to-Collision (TTC) failures drop by 62.3% and 62.4%, indicating that the planning-shaped future representation helps the model capture the motion trends of future traffic participants and thereby lowers potential collision risk. Driving Direction Compliance (DDC) failures also drop by 66.7%, further showing that future information improves the modeling of dynamic driving constraints. In contrast, Drivable Area Compliance (DAC) failures remain largely stable, indicating that the gains of PlanWAM concentrate on dynamic scene understanding and decision making. As shown in Fig. 5(b), the number of failures falls from 314 to 237, mainly from fewer trajectory scoring errors (−87). The predicted future latent thus provides effective foresight and improves the reliability of candidate-trajectory evaluation and selection. Proposal quality also improves to some extent, while proposal-collapse issues remain. This analysis indicates that, by learning a planning-oriented future representation, PlanWAM provides effective foresight for trajectory evaluation and selection, thereby reducing planning failures caused by insufficient modeling of future evolution.

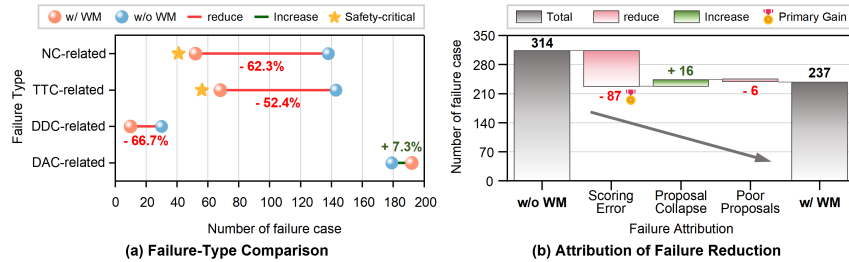


Figure 5: Analysis of failure reduction enabled by planning-oriented foresight. (a) Failure Type Distribution. (b) Attribution of Failure Reduction.

5 CONCLUSION

We presented PlanWAM, a planning-shaped world-action model that reformulates future modeling in end-to-end autonomous driving as planning-oriented predictive representation learning. By shaping future latent representations with trajectory planning objectives and distilling them from privileged future observations to a history-only prior, PlanWAM learns deployable foresight for trajectory generation and selection. PlanWAM achieves 93.8 PDMS / 90.9 EPDMS on NAVSIM-v1/v2 navtest and 38.7 HD-Score in zero-shot closed-loop HUGSIM evaluation, demonstrating the effectiveness of planning-shaped future representations for end-to-end autonomous driving.

REPRODUCIBILITY STATEMENT

The method of PlanWAM is specified in Sec. 3, including the Temporal Register Pyramid, the planning-shaped future posterior, Hindsight-to-Foresight Distillation, and the two-stage training and inference procedure. Dataset splits, evaluation protocols, and the main implementation settings are given in Sec. 4.1. Appendix A records the architecture, optimization, loss terms, and the PDMS / EPDMS / HUGSIM scoring protocols used for the reported numbers. All experiments use the public NAVSIM-v1, NAVSIM-v2, and HUGSIM benchmarks. We will submit anonymized source code with the supplementary material.

AI USE STATEMENT

In this work, we used generative AI tools to assist with translation between Chinese notes and English prose, to draft and edit parts of the manuscript for readability, to format tables and $\LaTeX$, and to assist with software implementation and debugging. We have not used generative AI tools to generate synthetic datasets, to formulate mathematical claims or proofs, or to invent experimental results, and these uses are not applicable to this work. All AI-assisted text, equations, tables, and code were reviewed by the authors; numerical results were checked against the original experimental logs. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

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A ADDITIONAL IMPLEMENTATION AND EVALUATION DETAILS

This appendix records the implementation and evaluation settings used for the results in the main paper. Inference never uses the privileged posterior: the deployed model encodes history, predicts $\hat{z}_t^+$ with the causal prior, and plans from that foresight.

A.1 ARCHITECTURE AND OPTIMIZATION

Table 9 lists the settings shared by all reported PlanWAM runs. Historical inputs are four frames from the front, left, right, and rear cameras. The training-only posterior reads future frames at 1-s intervals over the same 4-s horizon used for planning. Each candidate trajectory has eight waypoints at 0.5-s intervals. Scene registers come from a DINOv2 ViT-S backbone whose base weights stay frozen; only rank-32 LoRA adapters are trained. The Temporal Register Pyramid uses the recency-aware query allocation [4, 4, 8, 16] per camera, which yields the 128-token history h_t . The prior and posterior future latents both contain $N_z=256$ tokens, ordered by future time, camera, and register slot so that $\mathcal{L}_{\text{latent}}$ aligns corresponding tokens.

Table 9: **Implementation settings used for all reported PlanWAM results.**

Item	Setting
History	$H=4$ frames, four cameras
Future horizon (training posterior / planning)	4 s
Future observation stride (posterior only)	1 s
Waypoints	8, interval 0.5 s
Scene registers	16 per camera-frame
Visual backbone	DINOv2 ViT-S, frozen base, LoRA rank 32
History tokens	[4, 4, 8, 16] queries per camera, 128 tokens
Future latent	$N_z=256$ tokens
Trajectory candidates	$K=64$
Optimizer	AdamW
Hardware	16× NVIDIA A800, batch size 8 per GPU
Learning rate	Stage 1: 1×10^{-4} ; Stage 2: 1×10^{-5}
Latent-alignment weight	$\lambda_{\text{latent}}=0.05$

A.2 TRAINING OBJECTIVES

Stage 1 optimizes only the posterior branch with

$$\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{plan}} + \mathcal{L}_{\text{score}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}}. \quad (21)$$

$\mathcal{L}_{\text{plan}}$ is the minimum L_1 distance between the ground-truth trajectory and the K proposals. $\mathcal{L}_{\text{score}}$ is binary cross-entropy between the predicted subscores and the PDM targets of those proposals. $\mathcal{L}_{\text{reg}}$ combines a proposal-norm penalty and a diversity penalty that discourages collapsed modes; each term is weighted by 1×10^{-4} . Stage 2 freezes the posterior, initializes compatible prior parameters from that checkpoint, and fine-tunes the history-only prior with

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{task}} + \lambda_{\text{latent}} \mathcal{L}_{\text{latent}}, \quad \lambda_{\text{latent}}=0.05, \quad (22)$$

where $\mathcal{L}_{\text{latent}}$ is the stop-gradient token alignment in Eq. equation 17.

A.3 EVALUATION PROTOCOLS

NAVSIM-v1 PDMS follows the official non-reactive protocol: background actors replay recorded futures, and the ego vehicle tracks the planned trajectory for 4 seconds Dauner et al. (2024). With the official weights,

$$\text{PDMS} = \text{NC} \times \text{DAC} \times \frac{5 \text{ TTC} + 5 \text{ EP} + 2 \text{ C}}{12}. \quad (23)$$

Driving-direction compliance is computed but has zero weight in PDMS. NAVSIM-v2 reports the Extended PDM Score on the same navtest split Cao et al. (2025). Relative to PDMS, EPDMS adds traffic-light compliance, lane keeping, history comfort, and extended comfort, and treats no-at-fault collision, drivable-area compliance, driving-direction compliance, and traffic-light compliance as multiplicative penalties. HUGSIM evaluation deploys the NAVSIM-trained prior without

simulator-specific fine-tuning and reports route completion together with HD-Score Zhou et al. (2026a).